

RESEARCH PAPER REVIEW & RESEARCH REFERENCES

BhoomiDrishti

Predictive Analytics for Early Detection of Land Acquisition Delays

Smart India Hackathon 2026
Problem Statement ID: 26017

Literature Review • Research Gap • System Mapping

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1. Purpose of the Literature Review

This literature review establishes the academic and technical foundation for BhoomiDrishti, a predictive decision-support system intended to identify potential land-acquisition delays early, explain the factors driving risk, prioritize critical cases, and support preventive administrative action.

The reviewed literature was selected to cover both the domain problem (land/ROW acquisition, compensation, legal and rehabilitation issues) and the technical approach (machine learning, explainable AI and risk prioritization).

2. Research Themes

Theme	Relevance
Land & ROW Acquisition	Research on acquisition duration, parcels, condemnation, ownership and access restrictions.
Indian Delay Factors	Evidence from Indian highway and infrastructure projects.
Compensation, Legal & R&R	Compensation adequacy, disputes, displacement, rehabilitation and grievance issues.
Machine Learning & AI	Predictive models for construction/project delay risk.
Explainable AI	SHAP and interpretable models that explain why risk is high.
Prioritization & Decision Support	Moving from prediction to probability-impact ranking and action.

3. Summary of 20 Research Papers

No.	Research Paper	Year	Main Area	BhoomiDrishti Relevance
1	Right-of-Way Acquisition Duration Prediction Model for Highway Construction Projects	2012	ROW prediction	ROW acquisition duration prediction; identifies measurable acquisition factors.
2	Empirical Study on the Key Drivers Affecting Durations for Right-of-Way Acquisition on Highway Projects	2014	ROW drivers	Identifies drivers affecting ROW acquisition duration.
3	Optimizing Transparency and Disclosure to	2013	ROW transparency	Links condemnation, transparency and

	Reduce Right-of-Way Acquisition Duration for Construction Projects in Mississippi			price disclosure with ROW duration.
4	Delay Analysis of Infrastructure Construction Projects in India	2025	India delay factors	Indian infrastructure delay factors; land acquisition is important for road projects.
5	Critical Factors Influencing the Performance of Highway Projects: An Empirical Evaluation	2022	Indian highway factors	Indian highway factors including land acquisition, encroachment and disputes.
6	Investigation into the Causes of Delay in Land Acquisition for PPP Projects in Developing Countries	2017	Land acquisition causes	Resettlement, political interference, land availability, institutional and legal delays.
7	The Impact of Land Acquisition Delays on Time and Cost Performance in Construction Projects	2025	Time/cost impact	Quantifies the effect of acquisition delays on project time and cost.
8	Machine Learning Algorithms for Construction Projects Delay Risk Prediction	2020	ML prediction	Supports ML-based construction delay-risk prediction.
9	Applied Artificial Intelligence for Predicting Construction Projects Delay	2021	AI/ensemble	Evaluates multiple AI/ML and ensemble approaches for delay prediction.
10	Prediction of Risk Delay in Construction Projects Using a Hybrid Artificial Intelligence Model	2020	Hybrid AI	Supports hybrid AI approaches for delay-risk prediction.
11	AI-Powered Delay Risk Prediction in Construction	2026	Indian AI	Recent Indian evidence for advanced AI-

	Projects: Leveraging Machine Learning, Deep Learning and Hybrid Models			based delay prediction.
12	Identifying and Predicting Delay Risks in Prefabricated Construction: An Explainable Ensemble Learning Approach	2025	Explainable AI	Explainable ensemble learning and SHAP for identifying delay drivers.
13	Predictive Assessment of Construction Delays on Schedule, Cost, and Quality Performance Using Machine Learning and Multi-Criteria Decision Analysis	2026	ML + MCDA	Combines ML with MCDA for delay- factor assessment and prioritization.
14	From Prediction to Prioritization: Automated Framework for Delay Risk Management in Highway Earthwork Projects	2026	Prediction + prioritization	Combines prediction, probability-impact prioritization and SHAP.
15	Measuring Inadequacy in Compensation for the Compulsory Acquisition of Land: Evidence from Bengaluru, India	2022	Compensation	Examines compensation adequacy in compulsory land acquisition.
16	Land Acquisition and Resettlement and Rehabilitation Issues – A Success Story Based on Sardar Sarovar (Narmada) Project	2014	R&R	Case study of acquisition, displacement, R&R and grievance redressal.

17	Strategies for Road Project Execution with Land Access Restrictions: A Systematic Review	2026	Land access	Reviews technical, project-management, social, institutional and legal responses to land-access restrictions.
18	A Study of Land Acquisition and Compensation Process for National Highway in Maharashtra	2023	Maharashtra	Maharashtra-specific acquisition, compensation, opposition and grievance issues.
19	The Predicament of Land Acquisition, Displacement and Resettlement: An Analysis of Indian Scenario	2023	Indian R&R	Indian legal, social, displacement and R&R challenges.
20	Land Acquisition Infrastructure Development and BOOT Model Adoption for Private Universities in India	2026	Indian land acquisition	Recent Indian evidence on documentation, land availability, costs and institutional barriers.

4. Detailed Research Review

1. Right-of-Way Acquisition Duration Prediction Model for Highway Construction Projects

Authors: Imad Aleithawe, R. Ralph Sinno, William H. McAnally

Year: 2012

Publisher/Source: ASCE

DOI: 10.1061/(ASCE)CO.1943-7862.0000450

Research relevance: ROW acquisition duration prediction; identifies measurable acquisition factors.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

2. Empirical Study on the Key Drivers Affecting Durations for Right-of-Way Acquisition on Highway Projects

Authors: Taehong Sohn, Marcelo Azambuja, James T. O'Connor, William J. O'Brien

Year: 2014

Publisher/Source: ASCE

DOI: 10.1061/(ASCE)ME.1943-5479.0000264

Research relevance: Identifies drivers affecting ROW acquisition duration.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

3. Optimizing Transparency and Disclosure to Reduce Right-of-Way Acquisition Duration for Construction Projects in Mississippi

Authors: Imad Aleithawe

Year: 2013

Publisher/Source: ASCE

DOI: 10.1061/(ASCE)LA.1943-4170.0000119

Research relevance: Links condemnation, transparency and price disclosure with ROW duration.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

4. Delay Analysis of Infrastructure Construction Projects in India

Authors: L. Pinky Devi, Sindhu

Year: 2025

Publisher/Source: Springer

DOI: 10.1007/s40030-025-00899-5

Research relevance: Indian infrastructure delay factors; land acquisition is important for road projects.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the broader evidence base for Indian infrastructure, acquisition and R&R risk analysis.

5. Critical Factors Influencing the Performance of Highway Projects: An Empirical Evaluation

Authors: Shumank Deep et al.

Year: 2022

Publisher/Source: MDPI

DOI: 10.3390/buildings12060849

Research relevance: Indian highway factors including land acquisition, encroachment and disputes.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the broader evidence base for Indian infrastructure, acquisition and R&R risk analysis.

6. Investigation into the Causes of Delay in Land Acquisition for PPP Projects in Developing Countries

Authors: Solomon Babatunde, Onaopepo Adeniyi, Oluwaseyi Awodele

Year: 2017

Publisher/Source: Emerald

DOI: 10.1108/JEDT-05-2016-0029

Research relevance: Resettlement, political interference, land availability, institutional and legal delays.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

7. The Impact of Land Acquisition Delays on Time and Cost Performance in Construction Projects

Authors: Topan, Rita Tahir Lopa, Bambang Bakri

Year: 2025

Publisher/Source: ETASR

DOI: 10.48084/etasr.13165

Research relevance: Quantifies the effect of acquisition delays on project time and cost.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the broader evidence base for Indian infrastructure, acquisition and R&R risk analysis.

8. Machine Learning Algorithms for Construction Projects Delay Risk Prediction

Authors: Ahmed Gondia, Ahmad Siam, Wael El-Dakhakhni, Ayman H. Nassar

Year: 2020

Publisher/Source: ASCE

DOI: 10.1061/(ASCE)CO.1943-7862.0001736

Research relevance: Supports ML-based construction delay-risk prediction.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the ML/AI prediction layer and the use of historical project data to estimate delay risk.

9. Applied Artificial Intelligence for Predicting Construction Projects Delay

Authors: Christian Egwim et al.

Year: 2021

Publisher/Source: Elsevier

DOI: 10.1016/j.mlwa.2021.100166

Research relevance: Evaluates multiple AI/ML and ensemble approaches for delay prediction.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the ML/AI prediction layer and the use of historical project data to estimate delay risk.

10. Prediction of Risk Delay in Construction Projects Using a Hybrid Artificial Intelligence Model

Authors: Various authors; see official journal record

Year: 2020

Publisher/Source: MDPI

DOI: 10.3390/su12041514

Research relevance: Supports hybrid AI approaches for delay-risk prediction.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the ML/AI prediction layer and the use of historical project data to estimate delay risk.

11. AI-Powered Delay Risk Prediction in Construction Projects: Leveraging Machine Learning, Deep Learning and Hybrid Models

Authors: Pramodini Sahu et al.

Year: 2026

Publisher/Source: Taylor & Francis

DOI: 10.1080/15623599.2026.2619678

Research relevance: Recent Indian evidence for advanced AI-based delay prediction.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the ML/AI prediction layer and the use of historical project data to estimate delay risk.

12. Identifying and Predicting Delay Risks in Prefabricated Construction: An Explainable Ensemble Learning Approach

Authors: Yishuai Tian, Yiran Chen, Yan Ning

Year: 2025

Publisher/Source: Elsevier

DOI: 10.1016/j.aei.2025.104219

Research relevance: Explainable ensemble learning and SHAP for identifying delay drivers.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports explainability, prioritization and decision-support features beyond a simple prediction score.

13. Predictive Assessment of Construction Delays on Schedule, Cost, and Quality Performance Using Machine Learning and Multi-Criteria Decision Analysis

Authors: Radhe Shyam, Sanjay Tiwari

Year: 2026

Publisher/Source: Springer

DOI: 10.1007/s40030-026-00989-y

Research relevance: Combines ML with MCDA for delay-factor assessment and prioritization.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports explainability, prioritization and decision-support features beyond a simple prediction score.

14. From Prediction to Prioritization: Automated Framework for Delay Risk Management in Highway Earthwork Projects

Authors: Mingeon Cho et al.

Year: 2026

Publisher/Source: Elsevier

DOI: 10.1016/j.autcon.2025.106588

Research relevance: Combines prediction, probability-impact prioritization and SHAP.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports explainability, prioritization and decision-support features beyond a simple prediction score.

15. Measuring Inadequacy in Compensation for the Compulsory Acquisition of Land: Evidence from Bengaluru, India

Authors: See official journal record for complete author list

Year: 2022

Publisher/Source: MDPI

DOI: 10.3390/land11050664

Research relevance: Examines compensation adequacy in compulsory land acquisition.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

16. Land Acquisition and Resettlement and Rehabilitation Issues – A Success Story Based on Sardar Sarovar (Narmada) Project

Authors: A. Mahendran, Vinay Kumar, Afroz Ahmad

Year: 2014

Publisher/Source: Indian Journals

DOI: —

Research relevance: Case study of acquisition, displacement, R&R and grievance redressal.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the broader evidence base for Indian infrastructure, acquisition and R&R risk analysis.

17. Strategies for Road Project Execution with Land Access Restrictions: A Systematic Review

Authors: Luis Mayo-Alvarez, Fabiola Pasapera-Trujillo

Year: 2026

Publisher/Source: MDPI

DOI: 10.3390/buildings16122431

Research relevance: Reviews technical, project-management, social, institutional and legal responses to land-access restrictions.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports the broader evidence base for Indian infrastructure, acquisition and R&R risk analysis.

18. A Study of Land Acquisition and Compensation Process for National Highway in Maharashtra

Authors: Renuka R. Purohit, Rahul S. Chaudhari, Ankit Saraswat

Year: 2023

Publisher/Source: WJAETS

DOI: 10.30574/wjaets.2023.9.1.0122

Research relevance: Maharashtra-specific acquisition, compensation, opposition and grievance issues.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

19. The Predicament of Land Acquisition, Displacement and Resettlement: An Analysis of Indian Scenario

Authors: Abhilash Arun Sapre, Saira Gori

Year: 2023

Publisher/Source: SAGE

DOI: 10.1177/00219096231179651

Research relevance: Indian legal, social, displacement and R&R challenges.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

20. Land Acquisition Infrastructure Development and BOOT Model Adoption for Private Universities in India

Authors: N. Senthil Kumar, Viral Pandya

Year: 2026

Publisher/Source: Springer

DOI: 10.1007/s44217-026-01780-8

Research relevance: Recent Indian evidence on documentation, land availability, costs and institutional barriers.

Application to BhoomiDrishti: This study contributes evidence for selecting acquisition/project variables, designing predictive risk models, explaining risk drivers, or prioritizing interventions, depending on its research focus.

System connection: Supports domain-feature engineering and the classification of land, compensation, legal/social, administrative and documentation-related risks.

5. Research-to-System Mapping

System Component	Research Support	How It Informs the System
Historical + Current Data	Papers 1, 2, 4, 5, 8, 9	Use project/acquisition attributes as predictive features.
Land & Ownership Risk	Papers 1, 5, 6, 18, 20	Capture title, ownership, survey, encroachment and land-availability indicators.
Compensation Risk	Papers 3, 15, 18	Track compensation status, disputes, valuation and payment-related risk.
Judicial & Social Risk	Papers 3, 5, 6, 19	Track litigation, objections, resistance, protests and R&R disputes.
Government & Administrative Risk	Papers 4, 6, 17, 20	Represent approvals, coordination, documentation and institutional barriers.
R&R Risk	Papers 6, 16, 19	Track affected families, rehabilitation and resettlement progress.
ML Delay Prediction	Papers 8, 9, 10, 11	Support model selection and predictive-risk architecture.
Explainable Prediction	Papers 12, 13, 14	Use explainability to identify major contributors to risk.
Risk Prioritization	Papers 13, 14	Rank projects/cases by predicted risk and potential impact.
Decision Support	Papers 12, 13, 14, 17	Convert predictions into prioritized administrative attention and preventive action.

6. Five-Category Delay Risk Framework

1. Land & Ownership

Unclear ownership, multiple owners/heirs, outdated records, mutation, survey/boundary issues, encroachment and title conflicts.

2. Compensation & Financial

Compensation disputes, valuation issues, delayed payment, beneficiary/account issues and compensation grievances.

3. Judicial & Social

Formal ownership litigation, compensation cases, court stays, appeals, objections, landowner resistance, protests and R&R disputes.

4. Government & Administrative

Notifications, approvals, documentation, verification, inter-department coordination and administrative response time.

5. External & Project Factors

Forest/environment clearances, utility shifting, ROW restrictions, alignment/design changes and law-and-order issues.

Important classification note: unclear ownership is not automatically a court case. It begins as a land-record/ownership risk. When it develops into a formal dispute, objection, litigation, stay or appeal, it becomes a judicial/legal risk. This progression can be valuable as an early-warning signal for predictive modeling.

7. Identified Research Gap

- Existing research demonstrates that ROW and land-acquisition duration can be modeled using measurable factors.
- Indian studies identify land acquisition, compensation, disputes, encroachment, R&R and administrative issues as important project risks.
- Construction research demonstrates the usefulness of machine learning for delay-risk prediction.
- Recent research shows that explainable AI can identify the factors responsible for predicted risk.
- Recent frameworks show that prediction can be extended to probability-impact prioritization.
- However, these strands are generally studied separately and are not consistently integrated into a single land-acquisition decision-support workflow tailored to Indian government administration.

Research opportunity addressed by BhoomiDrishti: integrate domain-specific land-acquisition indicators with ML prediction, explainable risk factors, prioritization, GIS-based visibility, alerts and what-if decision support.

8. BhoomiDrishti Research Foundation

The overall research-to-system flow is:

Research Evidence → Delay Factors → Data Features → ML Prediction → Risk Probability/Score → Explainable Factors → Priority Ranking → Recommended Action → Monitoring & Alerts

The system is designed as a predictive intelligence layer that complements existing government process-monitoring and land-acquisition platforms. Its purpose is to help authorities move from tracking issues after they occur toward anticipating where intervention may be required.

9. Key Research Insights

- Land/ROW acquisition duration is influenced by measurable acquisition and project characteristics.

- Compensation, ownership, disputes, R&R and administrative coordination can interact and create cascading delays.
- Land acquisition delays can affect overall project schedule and cost.
- Machine learning can model complex relationships among multiple project-risk variables.
- Prediction alone is not sufficient for administration; explaining the drivers of risk improves usability.
- Prioritization helps authorities focus limited administrative attention on the most critical cases.
- A useful early-warning chain is: unclear ownership → ownership conflict → formal dispute → litigation/stay → delayed compensation or possession.
- Therefore, the system should monitor both current-stage problems and early indicators that may develop into more severe delays.

10. Reference and Copyright Note

This document is a research-reference and literature-review artifact prepared for the BhoomiDrishti project. Original research papers remain the intellectual property of their respective authors and publishers.

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- Retain original author, publication, publisher and DOI information.
- Do not modify or present published research as original work.

Appendix A — DOI / Publisher Reference List

No.	Paper	DOI	Publisher/Source
1	Right-of-Way Acquisition Duration Prediction Model for Highway Construction Projects	10.1061/(ASCE)CO.1943-7862.0000450	ASCE
2	Empirical Study on the Key Drivers Affecting Durations for Right-of-Way Acquisition on Highway Projects	10.1061/(ASCE)ME.1943-5479.0000264	ASCE
3	Optimizing Transparency and Disclosure to Reduce Right-of-Way Acquisition Duration for Construction Projects in Mississippi	10.1061/(ASCE)LA.1943-4170.0000119	ASCE
4	Delay Analysis of Infrastructure Construction Projects in India	10.1007/s40030-025-00899-5	Springer
5	Critical Factors Influencing the Performance of Highway Projects: An Empirical Evaluation	10.3390/buildings12060849	MDPI
6	Investigation into the Causes of Delay in Land Acquisition for PPP Projects in Developing Countries	10.1108/JEDT-05-2016-0029	Emerald
7	The Impact of Land Acquisition Delays on Time and Cost Performance in Construction Projects	10.48084/etasr.13165	ETASR
8	Machine Learning Algorithms for Construction Projects Delay Risk Prediction	10.1061/(ASCE)CO.1943-7862.0001736	ASCE
9	Applied Artificial Intelligence for Predicting Construction Projects Delay	10.1016/j.mlwa.2021.100166	Elsevier
10	Prediction of Risk	10.3390/su12041514	MDPI

	Delay in Construction Projects Using a Hybrid Artificial Intelligence Model		
11	AI-Powered Delay Risk Prediction in Construction Projects: Leveraging Machine Learning, Deep Learning and Hybrid Models	10.1080/15623599.2026.2619678	Taylor & Francis
12	Identifying and Predicting Delay Risks in Prefabricated Construction: An Explainable Ensemble Learning Approach	10.1016/j.aei.2025.104219	Elsevier
13	Predictive Assessment of Construction Delays on Schedule, Cost, and Quality Performance Using Machine Learning and Multi-Criteria Decision Analysis	10.1007/s40030-026-00989-y	Springer
14	From Prediction to Prioritization: Automated Framework for Delay Risk Management in Highway Earthwork Projects	10.1016/j.autcon.2025.106588	Elsevier
15	Measuring Inadequacy in Compensation for the Compulsory Acquisition of Land: Evidence from Bengaluru, India	10.3390/land11050664	MDPI
16	Land Acquisition and Resettlement and Rehabilitation Issues – A Success Story Based on Sardar Sarovar (Narmada) Project	—	Indian Journals
17	Strategies for Road Project Execution with Land Access	10.3390/buildings16122431	MDPI

	Restrictions: A Systematic Review		
18	A Study of Land Acquisition and Compensation Process for National Highway in Maharashtra	10.30574/wjaets.2023.9.1.0122	WJAETS
19	The Predicament of Land Acquisition, Displacement and Resettlement: An Analysis of Indian Scenario	10.1177/00219096231179651	SAGE
20	Land Acquisition Infrastructure Development and BOOT Model Adoption for Private Universities in India	10.1007/s44217-026-01780-8	Springer